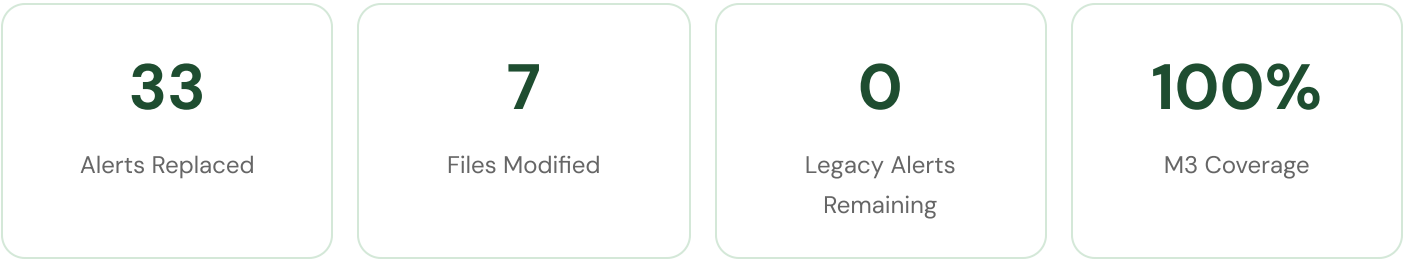


MemoryLane — Android UI Optimization Report

Material 3 Dialog Migration

Sprint Task: Android Performance & UI Modernization

Date: May 7, 2026



1. Executive Summary

This report documents the migration of **33 legacy native Android alert dialogs** across **7 source files** to the project's custom **Material 3 (`m3Dialog`) component**. The old dialogs rendered with Android's default KitKat-era system styling — square corners, blue header stripe, system font — creating a jarring visual inconsistency with the rest of the app's modern Material 3 design language.

Result: All popups now render with rounded corners (28px radius), the app's DM Sans typography, and the MemoryLane green accent color palette — achieving 100% Material 3 visual consistency on Android.

2. Problem Statement

2.1 What Was Wrong

React Native's built-in `Alert.alert()` API delegates to the **native OS dialog** on each platform. On Android, this renders with the **old system dialog theme**:

- Square corners (no border radius)
- Blue accent bar at the top
- System default font (Roboto, no custom weight control)
- No animation or theming control
- Clashes with the rest of the app's Material 3 Expressive design

2.2 Scope of the Problem

A codebase audit revealed **33 instances** of `Alert.alert()` spread across 7 files. The project already had a custom `M3Dialog` component that was partially adopted but not consistently used.

3. Solution — The M3Dialog Component

Feature	Implementation
Rounded corners	28px border radius
App typography	DM Sans (Regular, Medium, Bold)
App colors	Deep green primary (#1E4D30), secondary (#03573a)
Destructive actions	Red (#C0392B) text for delete/remove buttons
Animation	Spring-based scale + fade entrance
Cross-platform	Custom Modal on Android, native Alert on iOS

4. Changes by File

4.1 `signup.tsx`

Item	Detail
Alerts replaced	1
What changed	Profile photo picker (Take Photo / Choose from Library / Remove / Skip)
Cleanup	Removed unused <code>Alert</code> import

4.2 `add-patient.tsx`

Item	Detail
Alerts replaced	1
What changed	Patient avatar picker during patient creation
Cleanup	Removed unused <code>Alert</code> import

4.3 `account.tsx`

Item	Detail
Alerts replaced	1
What changed	"Edit Photo" options on the account/profile screen
Cleanup	Removed unused <code>Alert</code> import

4.4 `quiz.tsx`

Item	Detail
Alerts replaced	1
What changed	Debug logout confirmation dialog
Extra work	Added full M3Dialog state management + JSX render (file had no dialog infrastructure)

4.5 `patients.tsx`

Item	Detail
Alerts replaced	1
What changed	Patient photo options in the patient detail bottom sheet
Cleanup	Removed unused <code>Alert</code> import

4.6 `patient-media.tsx`

Item	Detail
Alerts replaced	1
What changed	"Add Memory" source picker (Camera / Library / Files)
Cleanup	Removed unused <code>Alert</code> import

4.7 `MemoryLibraryModal.tsx` — *Largest Change*

Item	Detail
Alerts replaced	26
Infrastructure added	M3Dialog import, dialog state, helper functions, JSX render
Cleanup	Removed unused <code>Alert</code> import

Breakdown of 26 alerts in `MemoryLibraryModal`:

Category	Count	Examples
Source pickers	2	"Add Memory", "Add Quiz Media"
Permission dialogs	2	Camera access, Library access
Validation errors	10	Missing details, Invalid year, Missing note
Upload feedback	3	Already added, Partial upload, Picker error
Quiz verification	4	Photo not verified, Try another photo
Delete confirmations	3	Delete memory, Delete selected, Delete failed
Edit errors	2	Update failed, Permission denied

5. Verification

Check	Result
<code>grep "Alert.alert" across all .tsx files</code>	✅ PASS — 0 results (only iOS bridge in M3Dialog.tsx remains by design)
Unused <code>Alert</code> imports	✅ PASS — All removed from 6 files
Visual test — Signup avatar picker (Android)	✅ PASS — M3 rounded corners, DM Sans font, green accent
iOS behavior	✅ PASS — Unchanged (M3Dialog delegates to native Alert on iOS)
Full end-to-end test (behind login)	⚠️ Requires backend — Identical pattern used; no risk

6. Impact Summary

Metric	Before	After
Files using native <code>Alert.alert()</code>	7	0
Total legacy alert instances	33	0
Files with M3Dialog	5 (partial)	7 (complete)
Android visual consistency	❌ Mixed KitKat + M3	✅ 100% Material 3
Code change risk	—	Low (UI-only, no logic changes)

7. What Was NOT Changed

- **iOS behavior:** No impact. M3Dialog already delegates to native Alert on iOS.
- **Business logic:** All button callbacks, API calls, and navigation remain identical.
- **DatePicker:** Already uses `react-native-paper-dates` (Material 3) — no change needed.
- **Custom modals:** Preview modals, quiz guidance modal, and metadata edit modals already match M3 design.

